

Trainings ▾

Preparatory Steps

- Remove the alternator drive belt. Refer to Procedure 013-005 in Section 13 (</qs3/pubsys2/xml/en/procedures/89/89-013-005.html>).
- Remove the alternator. Refer to Procedure 013-001 in Section 13 (</qs3/pubsys2/xml/en/procedures/89/89-013-001.html>).
- Remove the alternator drive gear and shaft. Refer to Procedure 009-009 in Section 9 (</qs3/pubsys2/xml/en/procedures/89/89-009-009.html>).
- Remove the cooling fan drive belt. Refer to Procedure 008-002 in Section 8 (</qs3/pubsys2/xml/en/procedures/89/89-008-002.html>).
- Remove the fan hub idler arm assembly. Refer to Procedure 008-002 in Section 8 (</qs3/pubsys2/xml/en/procedures/89/89-008-002.html>).
- Remove the vibration damper. Refer to Procedure 001-052 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-052.html>).
- Remove the gear cover and all related components. Refer to Procedure 001-031 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-031.html>).
- Remove both front idler gears. Refer to Procedure 001-097 (</qs3/pubsys2/xml/en/procedures/89/89-001-097.html>) and Refer to Procedure 001-098 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-098.html>).

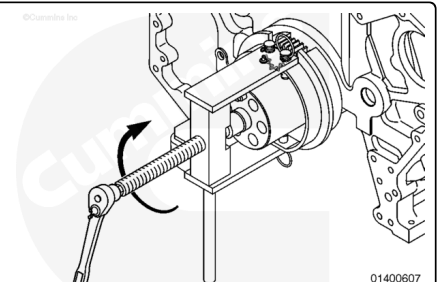
Remove

⚠ CAUTION ⚠

Heat can be used to aid the removal of the gear. Do not use a cutting torch. The high temperature of a torch will damage the teeth of the gear.

⚠ CAUTION ⚠

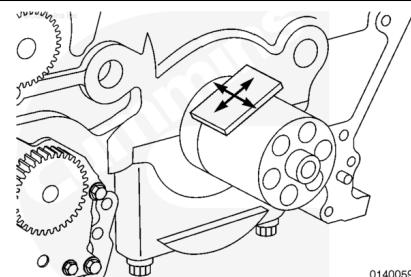
Do not exceed 475 N·m [350 ft-lb] of torque when turning the jackscrew.



Only remove the gear when the crankshaft or the gear is damaged.

Use a puller jaw, Part Number 3375835, and bridge assembly, Part Number 3165049, or equivalent, to remove the gear.

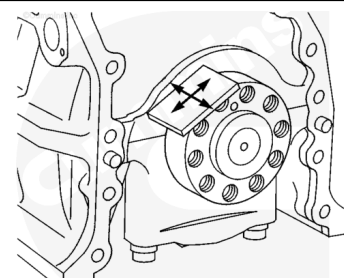
Use a hone stone to polish the shaft outside diameter. Remove all of the small scratches, burrs, and small grooves.



Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

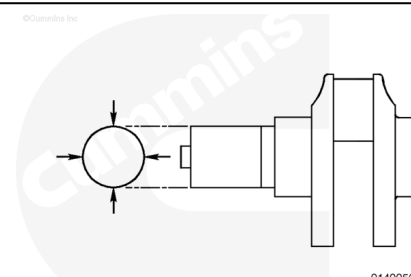
Use a hone stone and polish the crankshaft outside diameter.

Remove all of the small scratches, burrs, and small grooves.

Use care **not** to damage the crankshaft keyway.

Clean and inspect the crankshaft for reuse. Measure the outside diameter.

Crankshaft Outside Diameter (Gear Location) Front		
mm		in
112.13	MIN	4.415
112.15	MAX	4.416



Measure the inside diameter of the front crankshaft gear.

Front Crankshaft Gear Inside Diameter		
mm		in
112.030	MIN	4.411
112.055	MAX	4.412

If the crankshaft gear is **not** within specification, the crankshaft gear **must** be replaced.

Install

⚠ WARNING ⚠

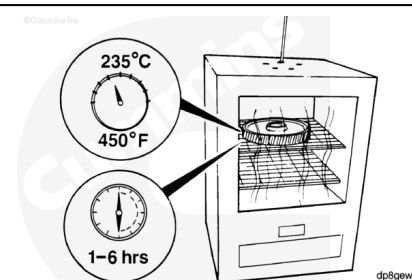
To reduce the possibility of severe burns, wear protective gloves when installing the heated ring gear.

⚠ CAUTION ⚠

Do not exceed the specified time or the temperature. Damage to the gear and the gear teeth will result.

⚠ CAUTION ⚠

Do not attempt to install the gear without using heat.



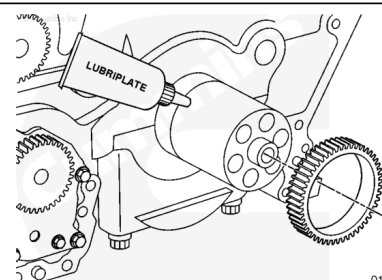
Use an oven and adjust the heat to 235°C [450°F]. Heat the gear for a minimum of one hour, but **not** more than six hours. The inside diameter of the gear will become larger and will simplify installation.

⚠ WARNING ⚠

To reduce the possibility of severe burns, wear protective gloves when installing the heated ring gear.

⚠ CAUTION ⚠

Allow the gear to air cool slowly. Do not use water or oil to reduce the cooling time. This will cause the gear to crack.



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Lubricate the outside diameter of the crankshaft with Lubriplate™ 105, or equivalent. Remove the gear from the oven and install it on the crankshaft.

Finishing Steps

- Install both front idler gears. Refer to Procedure 001-097 (</qs3/pubsys2/xml/en/procedures/89/89-001-097.html>) and Refer to Procedure 001-098 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-098.html>).
- Install the gear cover and all related components. Refer to Procedure 001-031 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-031.html>).
- Install the alternator drive gear and shaft. Refer to Procedure 009-009 in Section 9 (</qs3/pubsys2/xml/en/procedures/89/89-009-009.html>).
- Install the alternator. Refer to Procedure 013-001 in Section 13 (</qs3/pubsys2/xml/en/procedures/89/89-013-001.html>).
- Install the alternator drive belt. Refer to Procedure 013-005 in Section 13 (</qs3/pubsys2/xml/en/procedures/89/89-013-005.html>).
- Install the fan hub idler arm assembly. Refer to Procedure 008-002 in Section 8 (</qs3/pubsys2/xml/en/procedures/89/89-008-002.html>).
- Install the vibration damper. Refer to Procedure 001-052 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-052.html>).
- Install the cooling fan drive belt. Refer to Procedure 008-002 in Section 8 (</qs3/pubsys2/xml/en/procedures/89/89-008-002.html>).